Sina Moradi

RF / Microwave Engineer — PhD Researcher in Multiport RF Systems, Antenna Arrays, Noise, Photonics, and High-Frequency Modeling

Munich, Germany — Authorized to work in Germany — Open to relocation / international opportunities
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TUM Profile

Professional Summary

PhD researcher in High-Frequency Engineering at the Technical University of Munich with **thesis submission in progress**. My work spans **multiport RF systems, antenna arrays, beamforming, phased arrays, mutual coupling/matching, front-end mismatch, noise matching, correlated noise, and lossy transmission line effects**. I also have a strong background in **computational photonics**, including **anisotropic periodic multilayer structures, diffraction, LCoS / SLM modeling, and silicon photonics**. Additional experience includes **Cadence-based LNA/PA design, PCB design, RF measurements, OTA/chamber-based setups, and system-level simulation**. I am particularly interested in applied R&D roles in **RF systems and validation, phased-array / radar systems, high-frequency circuits, photonics / optical sensing, and advanced communication systems**.

Core Expertise

RF Systems & Arrays: multiport communication systems, antenna arrays, phased arrays, beamforming, mutual coupling, total active reflection coefficient (TARC), decoupling and matching networks, source/excitation optimization, end-to-end MIMO system thinking

Noise & Front-End Analysis: noise matching, correlated noise, front-end mismatching, lossy transmission-line noise, SNR / NF related analysis, receiver and transmitter chain behavior, high-impedance receiver concepts

Measurements & Validation: RF characterization, VNA / spectrum analyzer / signal generator / oscilloscope use, OTA and chamber-based setups, probe-station and near-field measurements / mmWave-style measurement exposure, calibration-sensitive workflows, structured post-processing and reporting

Circuits & Hardware: Cadence-based LNA/PA design, transistor-level RF simulation, PCB design, Altium Designer, KiCad, EMC-aware hardware implementation

Photonics & Wave Physics: anisotropic periodic multilayer structures, diffraction, Bloch–Floquet concepts, LCoS / SLM structures, silicon photonics, wave interaction in complex media

Additional Topics: reconfigurable intelligent surfaces (RIS), BAW filters, acousto-electromagnetic systems, radio-acoustic sensing concepts

Software & Simulation: MATLAB, Python, C++, Simulink, Keysight SystemVue, HFSS, CST, ADS, COMSOL, Altium Designer, KiCAD, FEKO, Momentum, Optenni Lab, Cadence, Lumerical, TCAD, LaTeX

Research & Professional Experience

Doctoral Research Assistant — **Oct 2020 – Present**
Technical University of Munich (TUM), Chair of High-Frequency Engineering — Munich, Germany

- Research on **multiport RF front-ends and antenna arrays**, including beamforming, matching, coupling reduction, scan-dependent performance, and source/excitation optimization.
- Investigated **noise matching, correlated noise, front-end mismatching, and lossy transmission-line effects** in high-frequency RF chains.
- Built quantitative **MATLAB/Python/SystemVue workflows** for RF system analysis, technical evaluation, and interpretation of measured/simulated data.
- Performed and supported **RF laboratory measurements**, calibration-sensitive evaluations, and reproducible documentation/post-processing workflows.

RF Engineer & J1 Visiting Faculty Researcher — **Oct 2024 – May 2025**
Virginia Tech, Bradley Department of Electrical & Computer Engineering — Blacksburg, VA, USA

- Conducted applied work on **front-end mismatching, mutual coupling, transmission-line noise, bandwidth, and Tx/Rx system trade-offs**.
- Worked with **Keysight SystemVue** on **end-to-end MIMO RF/baseband system modeling**, from transmit chain through physical channel to receiver.
- Identified and reported a **SystemVue modeling issue** to Keysight and discussed it directly with their engineers/developers.

Graduate Researcher — **Dec 2018 – Sep 2020**
University of Calgary — Calgary, Canada

- Designed, fabricated, and measured circularly polarized microstrip patch antennas.
- Built strong practical experience in **RF prototyping, antenna measurement, and network analysis**.

Selected Research Themes

- **Beamforming / phased arrays / multiport systems:** excitation-aware and excitation-independent characterization of coupled arrays; TARC, matching, scan behavior, and channel/system-level insight.
- **Physical elimination of decoupling and matching networks:** source-side / pre-distortion style realization of matching behavior in multiport antenna arrays.
- **Photonics / optical modeling:** M.Sc. thesis on **diffraction in multi-layer anisotropic periodic structures**, with relevance to LCoS / SLMs and wave-based sensing systems.
- **Advanced device/system topics:** RIS, BAW filters, acousto-electromagnetic systems, radio-acoustic sensing, silicon photonics, and RF–photonics-adjacent concepts.

Education

PhD Researcher, High-Frequency Engineering — **2020 – Present**
Technical University of Munich — Munich, Germany

- Research focused on RF systems, antenna arrays, multiport networks, noise, beamforming, and high-frequency modeling.

M.Sc. Electrical Engineering (Communications – Field & Waves) — **2016 – 2018**
University of Tehran — Tehran, Iran

- GPA: 18.27/20 — **Ranked 1st in the whole Faculty of Electrical & Computer Engineering**.
- Thesis: *Analysis of Diffraction by Multi-Layer Anisotropic Periodic Structures*.

B.Sc. Electrical Engineering, Communication Systems — **2012 – 2016**
Shahid Beheshti University — Tehran, Iran

- GPA: 19.34/20 — **Ranked 1st in the whole Faculty of Electrical & Computer Engineering**.

Selected Publications & Manuscripts

- S. Moradi, B. Honarbakhsh, and T. F. Eibert, "Some Excitation Independent Bounds for the Total Active Reflection Coefficient of Antenna Arrays," *IEEE Transactions on Antennas and Propagation*, 2022.
- S. Moradi, B. Honarbakhsh, and T. F. Eibert, "Reply to the Comments on 'Some Excitation Independent Bounds for the Total Active Reflection Coefficient of Antenna Arrays'," *IEEE Transactions on Antennas and Propagation*, 2023.
- M. Manteghi and S. Moradi, "Front-End Mismatching, Mutual Coupling, Bandwidth, Transmission Line Noise, and SNR," *EuCAP*, 2024.
- S. Moradi, E. Mehrshahi, and J. A. Nossek, "Physical Elimination of Decoupling and Matching Networks in Multiport Antenna Arrays," *IEEE AP-S/URSI*, 2025.
- S. Moradi, E. Mehrshahi, T. F. Eibert, and J. A. Nossek, "Constrained Source Impedance and Excitation Optimization for Beamforming in Mutually Coupled Antenna Arrays Without Decoupling Networks," *under review at IEEE Transactions on Antennas and Propagation*, 2026.

Honors & Additional Training

- Ranked 1st in both B.Sc. and M.Sc. studies
- Direct M.Sc. admission as brilliant talent; also admitted to Sharif University of Technology under brilliant-talent route before choosing the University of Tehran track
- National Elite Foundation member
- Top 0.1% in the Iranian nationwide undergraduate entrance exam
- **Silicon Photonics Design, Fabrication and Data Analysis**, Sharif University of Technology
- Intermediate C++ Programming certificate

Languages

Persian (native), English (C2 / TOEFL iBT), German (A2)

Technische Universität München | Arcisstraße 21 | 80333 München

To Whom It May Concern

München, 4. August 2025

Letter of Reference for Mr. Sina Moradi

It is with great pleasure that I write this letter of recommendation for Mr. Sina Moradi, born May 11, 1994 in Tehran, Iran, who has been performing doctoral research under my supervision at the Chair of High-Frequency Engineering, Technical University of Munich (TUM), since October 2020.

The research contributions of Mr. Moradi span from RF systems, mutual coupling reduction, and antenna array beamforming to noise correlation modeling in mismatched cascaded front-end receiving/transmitting chains—all of which are highly relevant for modern communication systems such as massive MIMO, 5G/6G and mmWave transceivers.

Beyond radio-frequency systems, Mr. Moradi has shown remarkable depth in computational photonics and the modeling of multi-layer periodic anisotropic structures. His Master's research on liquid crystal on silicon spatial light modulators (LCoS SLMs) and optical diffraction is highly regarded. In our lab, Mr. Moradi is known not only for his deep theoretical insight and advanced modeling skills (MATLAB, HFSS, CST, ADS, SystemVue, Lumerical, COMSOL, Feko), but also for his ability to solve complex problems independently and creatively. His research results have been published in leading IEEE journals and international conferences, including IEEE Transactions on Antennas and Propagation, EuCAP, and IEEE AP-S/URSI.

Moreover, Mr. Moradi demonstrates excellent soft skills, which are essential for success in both academic and industrial environments. He has a natural talent for problem solving, team management, professional networking, and interdisciplinary collaboration. He frequently acts as a bridge between people and ideas—something I have personally observed in both international projects and internal group work.

As part of his broad engagement, Mr. Moradi successfully initiated and completed a 6-month international research collaboration as a J1 Visiting Faculty Member at Virginia Tech (USA), where he worked closely with faculty in the field of RF system modeling and antenna array design. This initiative showcased his ability to connect across institutions and cultures and to drive high-impact research forward. Additionally, Mr. Moradi has actively collaborated with other chairs at TUM, including the Chair of Circuit Design and Signal Processing, where he worked with Professor Josef Nossek on advanced signal processing and multiport antenna systems.

What truly sets Mr. Moradi apart is his rare combination of high-level academic performance and sustained professional athletic involvement. Despite the intensity of his doctoral research, he has competed at semi-professional and league levels in table tennis, soccer, and CrossFit, both in Germany and Canada. He also holds an official coaching certificate from Germany and has

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School of Computation, Information and Technology
Department of Electrical Engieering
Chair of High-Frequency Engineering

Prof. Dr.-Ing Thomas Eibert

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coached youth athletes. Balancing this demanding academic and athletic lifestyle with discipline and energy is something I have rarely seen at the doctoral level.

In summary, I give Mr. Moradi my full recommendation for any advanced research, R&D, or engineering position in RF, photonics, applied electromagnetics, or system design. His technical breadth, personal discipline, and collaborative mindset make him an outstanding candidate for any environment.

Please feel free to contact me should you need any further information.

Sincerely,

Prof. Dr.-Ing. Thomas Eibert